

Week 3

Decision Tree

Outline

- Decision Tree- basic concepts
- Decision Tree – Example
- Decision Tree – Evaluation of Model

Key Problems of ML: Data Classification and Prediction

- Data classification
 - Classification → prediction
- Methods of classification
 - decision tree induction
 - Bayesian classification
 - backpropagation
 - association rule mining



What are Decision Trees?

Type of Supervised Learning Algorithm

Mostly used in Classification Problems

Works for both categorical and continuous data (I/O variables)

Nodes on each branch represents a choice between a number of alternatives

Each leaf node represents a decision

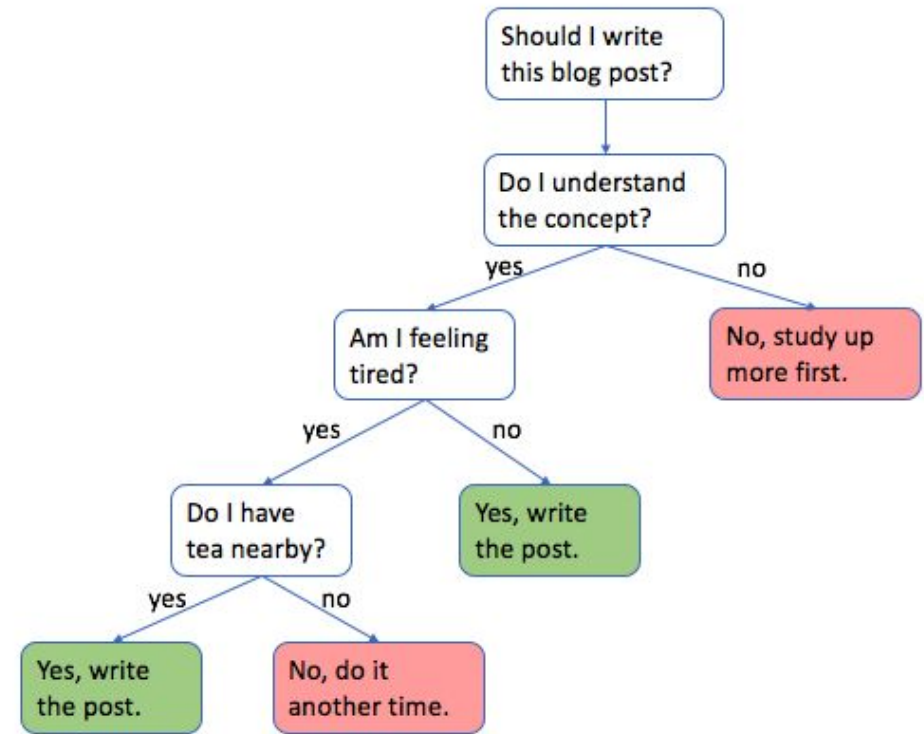
- Presentation form: "If, t hen" statements (Rules)
- Graphically – These look like trees

What Do these Trees Look Like?

- Works like a flow chart
- Captured as an upside-down tree
- Nodes
 - appear as rectangles or circles
 - represent test or decision
- Lines or branches - represent outcome of a test
- Circles - terminal (leaf) nodes
- Top or starting node- root node
- Internal nodes - rectangles

Basic Concepts

- A decision tree is a decision support tool
- For classification
- For regression (with quantification)
- It uses a tree – with nodes and branches
- Payoff table



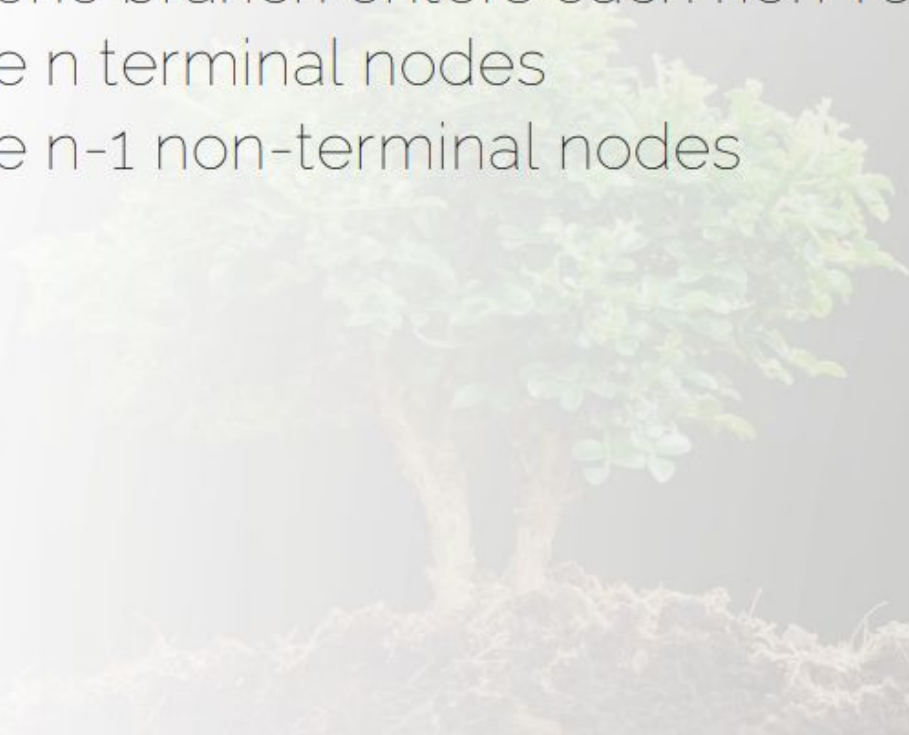
Alternative	State of Nature		
	Good Market(\$)	Average Market (\$)	Poor Market (\$)
Construct a large plant	100,000	35,000	-60,000
Construct a small plant	75,000	25,000	-40,000
Do nothing	-5,000	0	0

Binary Trees

- Characteristics
 - two branches leave each non-terminal node
 - those two branches cover outcomes of the test
 - exactly one branch enters each non-root node
 - there are n terminal nodes
 - there are $n-1$ non-terminal nodes

Nonbinary Trees

- Characteristics
 - two or more branches leave each non-terminal node
 - those branches cover outcomes of the test
 - exactly one branch enters each non-root node
 - there are n terminal nodes
 - there are $n-1$ non-terminal nodes



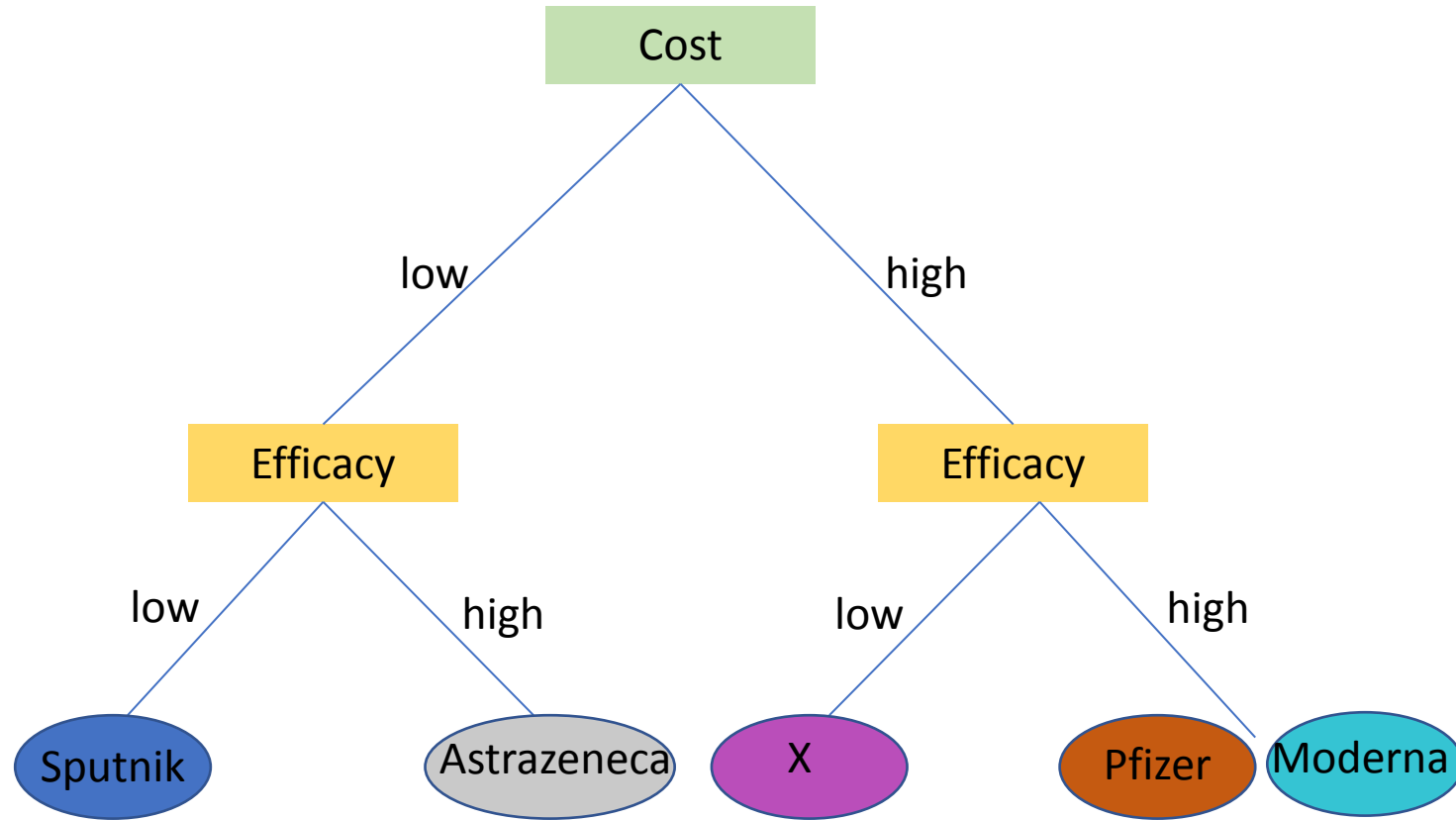
Example 1: COVID Vaccine

Let us classify some of the COVID vaccines: Pfizer, Moderna, Astrazeneca, Sputnik, and X. Pfizer, Moderna, and X vaccines are very costly while Astrazeneca and Sputnik are cheap. On the other hand, efficacies for Pfizer, Moderna and Astrazeneca are high while Sputnik and X are low. Draw a payoff table based on this information.

Payoff Table

Type of Vaccines	Cost and Benefit	
	Cost	Efficacy
Pfizer	High	High
Moderna	High	High
Astrazeneca	Low	High
Sputnik	Low	Low

Decision Tree

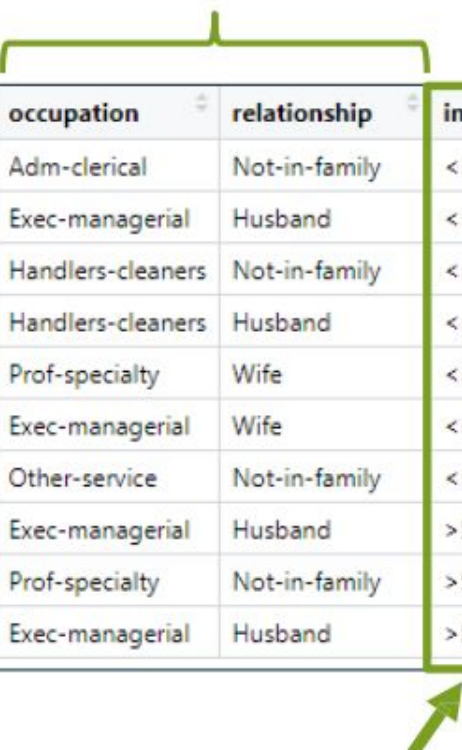


Trees Are Rules

Classification trees embody a series of rules to assign/predict *labels*.

Take the following sample of the *Adult Census* data...

Feature/Variables



	occupation	relationship	income
1	Adm-clerical	Not-in-family	<=50K
2	Exec-managerial	Husband	<=50K
3	Handlers-cleaners	Not-in-family	<=50K
4	Handlers-cleaners	Husband	<=50K
5	Prof-specialty	Wife	<=50K
6	Exec-managerial	Wife	<=50K
7	Other-service	Not-in-family	<=50K
8	Exec-managerial	Husband	>50K
9	Prof-specialty	Not-in-family	>50K
10	Exec-managerial	Husband	>50K

Labels

Using the data to the left, consider the following "rules":

IF **occupation** IS IN ("Adm-clerical", "Handlers-cleaners", "Other-service") THEN **income** = "<=50K"

ELSE IF **relationship** = "Wife" THEN **income** = "<=50K"

ELSE **income** = ">50K"

How does the classification tree algorithm arrive at these rules?

Classification Trees Minimize "Impurity"

Machine learning algorithms work by trying to achieve an *objective*.

In the case of classification trees, the objective is to *minimize impurity*.

Feature Values

Label Counts

	occupation	LTE50K	GT50K
1	Adm-clerical	1	0
2	Exec-managerial	2	2
3	Handlers-cleaners	2	0
4	Other-service	1	0
5	Prof-specialty	1	1

These are "pure"

Feature Values

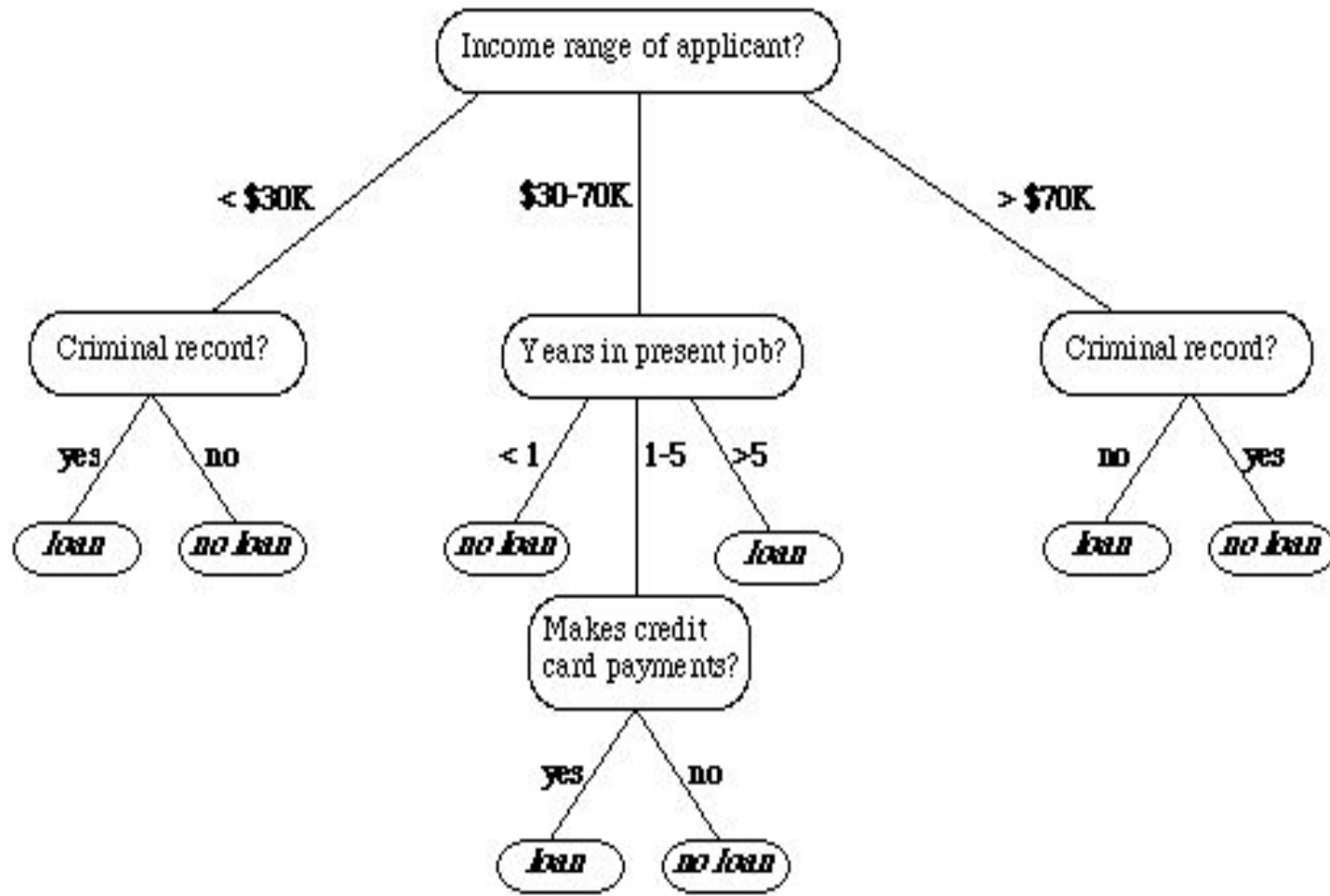
Label Counts

	relationship	LTE50K	GT50K
1	Husband	2	2
2	Not-in-family	3	1
3	Wife	2	0

These are "impure"

The classification tree algorithm iteratively uses the features to split the data into the largest collections of "purest" labels possible.

Example 2 – Loan Application



What are the rules?

Example 3: Restaurant (Russell and Norvig, 2003)

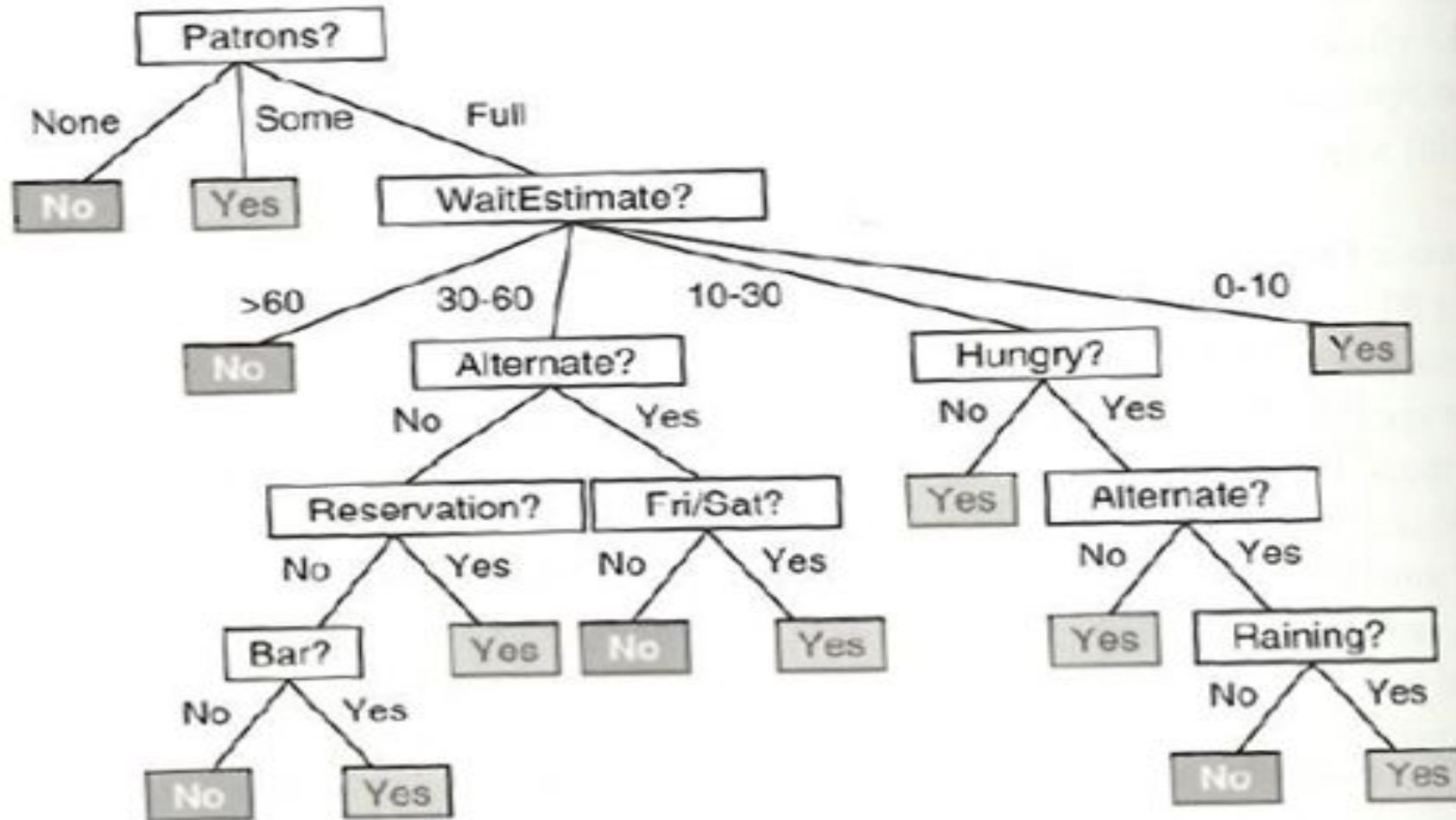
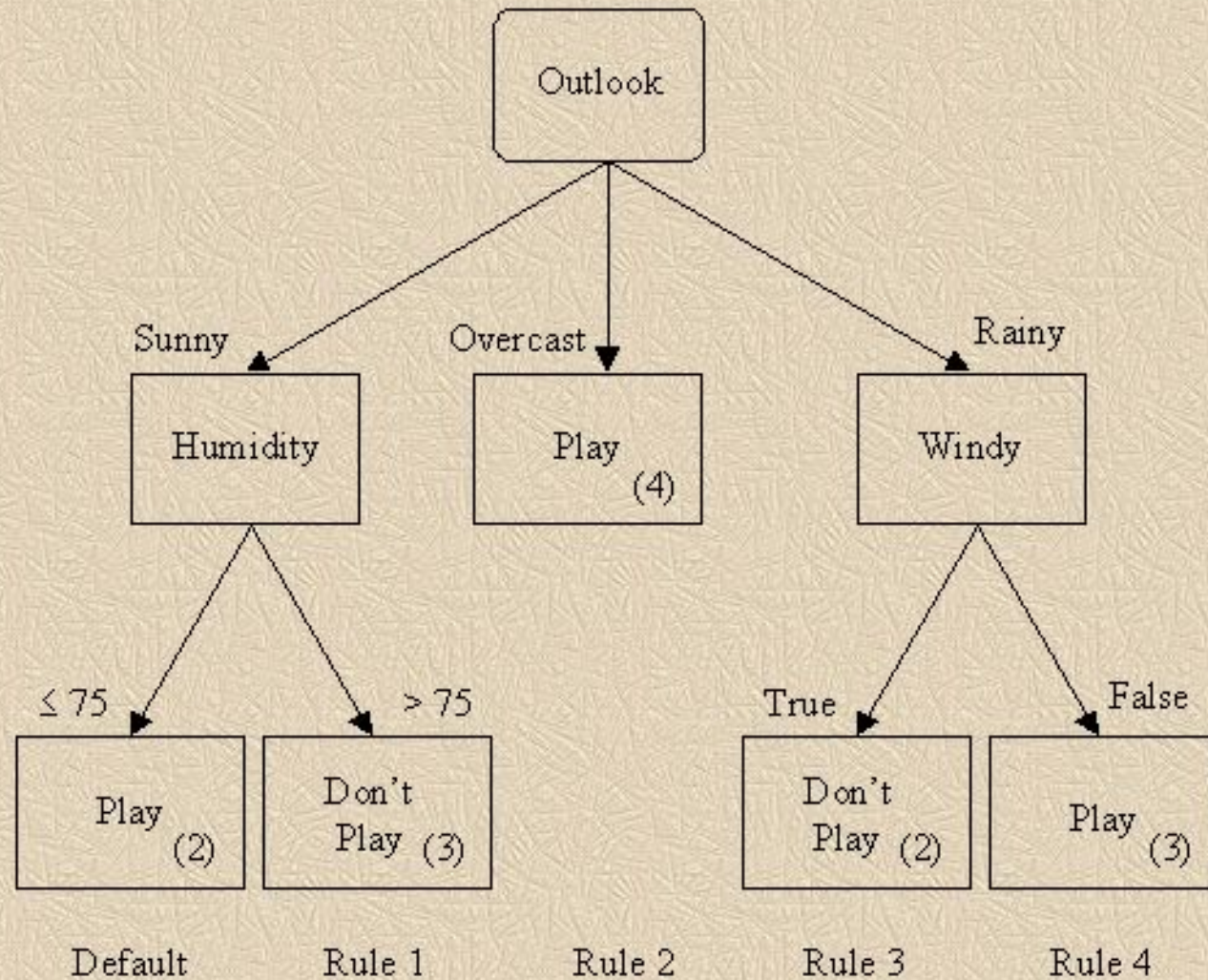


Figure 18.2 A decision tree for deciding whether to wait for a table.

Data for Golf

<i>Outlook</i>	<i>Temperature</i>	<i>Humidity</i>	<i>Windy</i>	<i>Play (positive) / Don't Play (negative)</i>
sunny	85	85	false	Don't Play
sunny	80	90	true	Don't Play
overcast	83	78	false	Play
rain	70	96	false	Play
rain	68	80	false	Play
rain	65	70	true	Don't Play
overcast	64	65	true	Play
sunny	72	95	false	Don't Play
sunny	69	70	false	Play
rain	75	80	false	Play
sunny	75	70	true	Play
overcast	72	90	true	Play
overcast	81	75	false	Play
rain	71	80	true	Don't Play

Example 4: Play Golf



What are the rules?

Example 5: Freight Charges - Transcript

Air shipping charges are set depending on the weight of a parcel. The basic rate is \$3/lb, reducing to \$2/lb for excess over 20 lb, with a minimum of \$6. Surface freight is \$2/lb for express delivery. However, this rate only applies in the local delivery area. If the shipping address is outside the local area and the parcel weighs more than 20lb, or express delivery is not required, the surface rate is the same as for local delivery (express). Normal delivery of packages is \$2/lb, while express delivery up to 20lb is \$2/lb, with \$1 express surcharge (per pound).

Notwithstanding the provisions of the previous paragraph, air freight to destinations west of the Mississippi is charged at double rate”

Courtesy: <http://www.cs.toronto.edu/~jm/340S/Slides6/Rules.pdf>

Extracted Information

Air shipping charges are set depending on the weight of a parcel. The basic rate is **\$3/lb**, reducing to **\$2/lb for excess over 20 lb**, with a **minimum of \$6**. **Surface freight** is **\$2/lb** for **express delivery**. However, this rate only applies in the **local delivery area**. If the shipping address is outside the **local area** and the parcel **weighs more than 20lb**, or **express delivery** is not required, the surface rate is the **same as for local delivery (express)**. Normal delivery of packages is **\$2/lb**, while **express delivery up to 20lb** is **\$2/lb**, with **\$1 express surcharge (per pound)**.

Notwithstanding the provisions of the previous paragraph, **air freight** to destinations **west of the Mississippi** is charged at **double rate**”

Air Shipping

Basic Rate - \$3/lb (2lb < weight ≤ 20lb)

Minimum Rate - \$6 (2 lb or less)

Excess over 20 lb - \$2/lb (> 20lb)

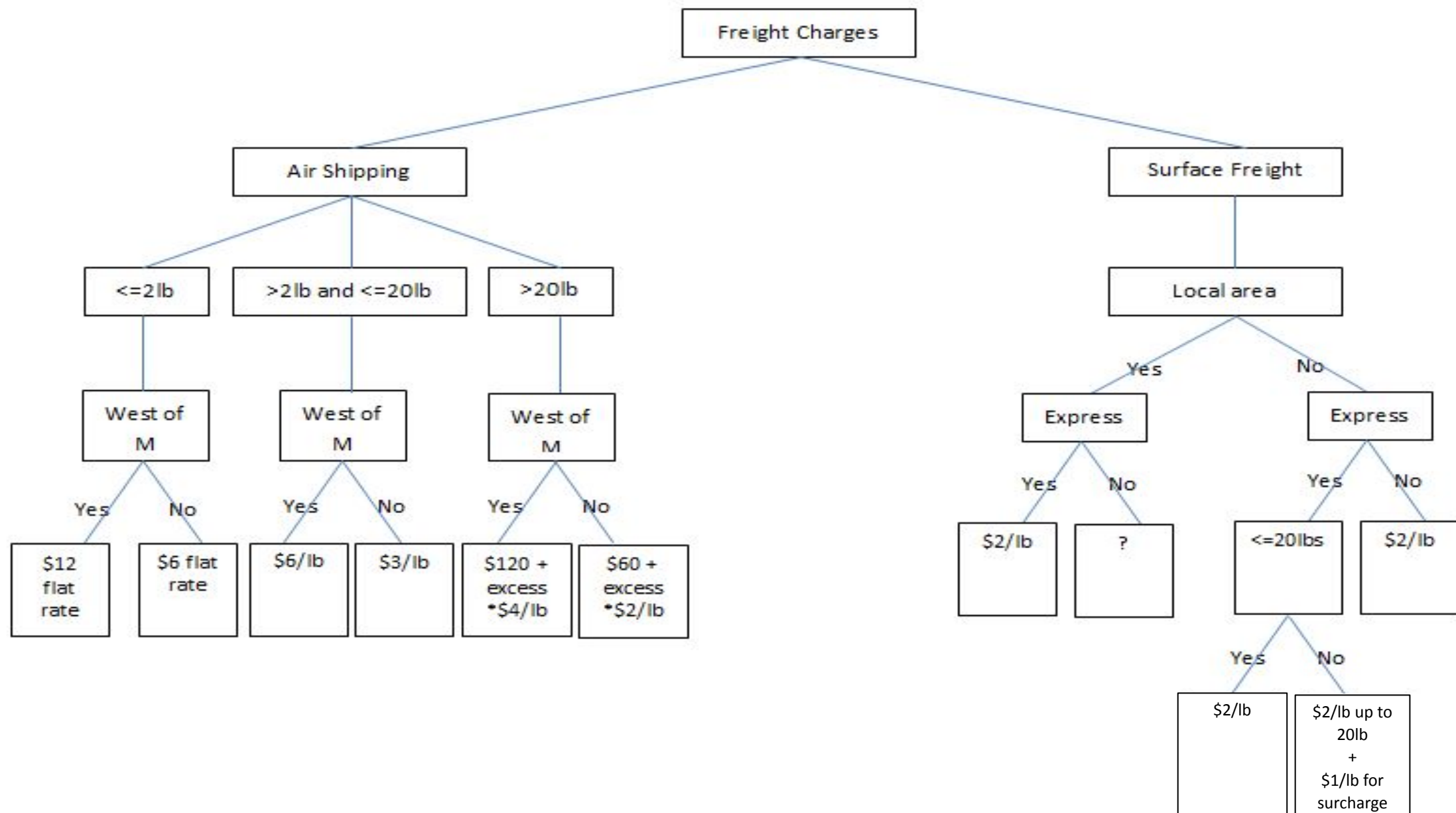
West of Mississippi – double rate

Surface Freight

Express Delivery, **local area** - \$2/lb

Outside local area

- no express delivery, > 20lb - \$2/lb
- Normal delivery - \$2/lb
- Express delivery, > 20lb - \$2/lb up to 20lb + \$1/lb



Example 6: Old Wildcatter Problem

Decision – to drill or not to drill?

Probability of well condition
(sum of probabilities = 1) :

- Dry with probability 0.5
- Wet with probability 0.3
- Soaking with probability 0.2

Monetary Payoffs of Oil Wildcatter's Problem

	Drill	Not Drill
Dry	0	0
Wet	\$700,000	0
Soaking	\$1,200,000	0

Cost of drilling (sum of probabilities = 1):

- Low, with probability 0.2, (\$150,000)
- Fair, with probability 0.6 (\$300,000)
- High, with probability 0.2 (\$500,000)

All in 1 table

Well Condition	Probability
Dry	0.5
Wet	0.3
Soaking	0.2

Cost of Drilling		
	Probability	Cost
Low	0.2	\$150,000
Fair	0.6	\$300,000
High	0.2	\$500,000

Monetary Payoffs		
Well Condition	Drill	Not Drill
Dry	0	0
Wet	\$700,000	0
Soaking	\$1,200,000	0

Observation	Drilling Cost	Well Condition	Drill?	Formula X-Y	Value
1	Low	Dry	Yes		
2	Low	Dry	No		
3	Low	Wet	Yes		
4	Low	Wet	No		
5	Low	Soaking	Yes		
6	Low	Soaking	No		
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					

Payoffs	Drill	No Drill
Dry	0	0
Wet	\$700,000	0
Soaking	\$1,200,000	0
Drilling Cost		
Low	\$150,000	
Fair	\$300,000	
High	\$500,000	

Observation	Drilling Cost	Well Condition	Drill?	Formula	Value
1	Low	Dry	Yes	= 0 - \$150,000	-\$150,000.00
2	Low	Dry	No		\$0.00
3	Low	Wet	Yes	= \$700,000 - \$150,000	\$550,000.00
4	Low	Wet	No		\$0.00
5	Low	Soaking	Yes	= \$1,200,000 - \$150,000	\$1,050,000.00
6	Low	Soaking	No		\$0.00
7	Fair	Dry	Yes	= 0 - \$300,000	-\$300,000.00
8	Fair	Dry	No		\$0.00
9	Fair	Wet	Yes	= \$700,000 - \$300,000	\$400,000.00
10	Fair	Wet	No		\$0.00
11	Fair	Soaking	Yes	= \$1,200,000 - \$300,000	\$900,000.00
12	Fair	Soaking	No		\$0.00
13	High	Dry	Yes	= 0 - \$500,000	-\$500,000.00
14	High	Dry	No		\$0.00
15	High	Wet	Yes	= \$700,000 - \$500,000	\$200,000.00
16	High	Wet	No		\$0.00
17	High	Soaking	Yes	= \$1,200,000 - \$500,000	\$700,000.00
18	High	Soaking	No		\$0.00

Calculation

Drill

- Low - total is \$300,000
 - dry: $0.5 * (\$-150,000) = -\$75,000$
 - wet: $0.3 * (\$550,000) = \$165,000$
 - soaking: $0.2 * (\$1,050,000) = \$210,000$
- Fair – total is \$150,000
 - dry: $0.5 * (\$-300,000) = -\$150,000$
 - wet: $0.3 * (\$400,000) = \$120,000$
 - soaking: $0.2 * (\$900,000) = \$180,000$
- High – total is -\$50,000
 - dry: $0.5 * (\$-500,000) = -\$250,000$
 - wet: $0.3 * (\$200,000) = \$60,000$
 - soaking: $0.2 * (\$700,000) = \$140,000$

Calculation for No Drill?

Final Calculation

Drill – total is \$140,000

- Low – $0.2 * (\$300,000) = \$60,000$
- Fair - $0.6 * (\$150,000) = \$90,000$
- High - $0.2 * (-\$50,000) = -\$10,000$

No Drill?

Draw the Decision Tree

Drill

- Low - total is \$300,000

- dry: $0.5 * (\$-150,000) = -\$75,000$
- wet: $0.3 * (\$550,000) = \$165,000$
- soaking: $0.2 * (\$1,050,000) = \$210,000$

- Fair – total is \$150,000

- dry: $0.5 * (\$-300,000) = -\$150,000$
- wet: $0.3 * (\$400,000) = \$120,000$
- soaking: $0.2 * (\$900,000) = \$180,000$

- High – total is -\$50,000

- dry: $0.5 * (\$-500,000) = -\$250,000$
- wet: $0.3 * (\$200,000) = \$60,000$
- soaking: $0.2 * (\$700,000) = \$140,000$

Drill – total is \$140,000

- Low – $0.2 * (\$300,000) = \$60,000$
- Fair - $0.6 * (\$150,000) = \$90,000$
- High - $0.2 * (-\$50,000) = -\$10,000$

Oil Wildcatter's Problem

The DTREE Procedure Line-printer Plot

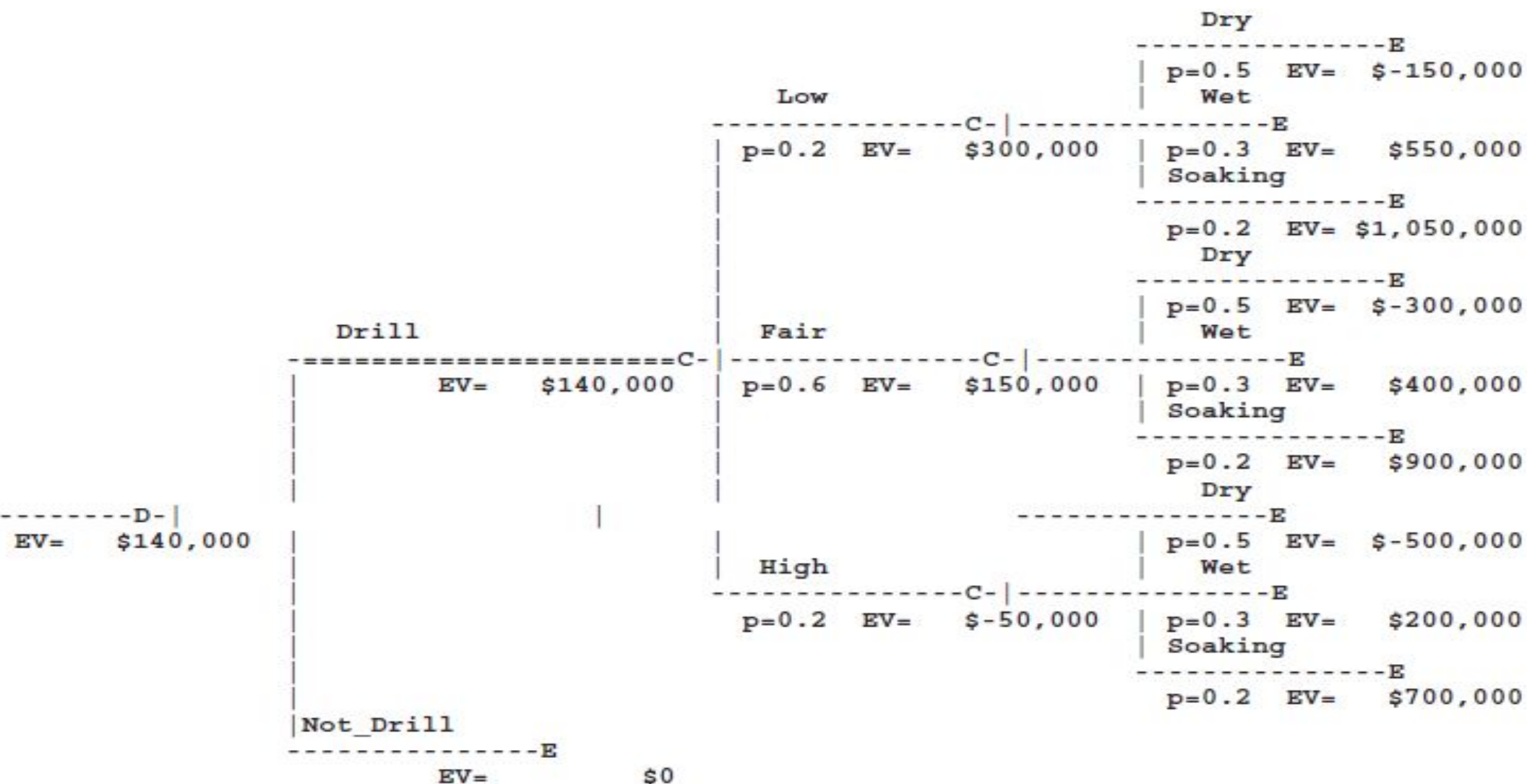


Figure 3.3. The Decision Tree

Evaluation of Decision Tree Model

- Entropy
- Information Gain
- Gini index

Entropy

- Measure of **impurity**, disorder, or **uncertainty** in a set of examples
- Measure the randomness of information being processed
- The **higher the entropy value**, the **more difficult** to draw conclusions from that information
- Entropy controls how a decision tree splits the data
- <https://towardsdatascience.com/entropy-how-decision-trees-make-decisions-2946b9c18c8>

Entropy Formula

- <https://towardsdatascience.com/entropy-how-decision-trees-make-decisions-2946b9c18c8>
- <http://www.learnbymarketing.com/481/decision-tree-flavors-gini-info-gain/>
- <https://www.math.unipd.it/~aiolli/corsi/0708/IR/Lez12.pdf>
- $P(\text{value}_i)$ – probability of getting value_{*i*}

$$\text{entropy}(\text{Set}) = I(\text{Set}) = - \sum_{i=1}^k P(\text{value}_i) \cdot \log_2(P(\text{value}_i))$$

Example 1 - Entropy

$$\text{entropy}(\text{Set}) = I(\text{Set}) = -\sum_{i=1}^k P(\text{value}_i) \cdot \log_2(P(\text{value}_i))$$

1.1 We have a set of items:

$R = \{z, z, z, z, z, y, y, y\}$

$P(z) = 5/8$

$P(y) = 3/8$

- Calculate the entropy of the set

- $\text{Entropy}(R) = I(R)$

$$= - [P(z) \times \log_2(P(z)) + P(y) \times \log_2(P(y))]$$

$$= - [5/8 \times \log_2(5/8) + 3/8 \times \log_2(3/8)]$$

$$= - [5/8 \times (-0.678) + 3/8 \times (-1.415)]$$

$$= - [-0.954]$$

$$= +0.954$$

1.2 We have a set of items:

$R = \{z, z, z, z, y, y, y, y\}$

- Calculate the entropy of the set

- $\text{Entropy}(R) = I(R)$

$$= - [P(z) \times \log_2(P(z)) + P(y) \times \log_2(P(y))]$$

$$= - [4/8 \times \log_2(4/8) + 4/8 \times \log_2(4/8)]$$

$$= - [4/8 \times (-1) + 4/8 \times (-1)]$$

$$= - [-1]$$

$$= +1$$

(Maximum impurity) – good for training

1.3 We have a set of items:

$R = \{y, y, y, y, y, y, y, y\}$

- Calculate the entropy of the set

- $\text{Entropy}(R) = I(R)$

$$= - [P(z) \times \log_2(P(z)) + P(y) \times \log_2(P(y))]$$

$$= - [0/8 \times \log_2(0/8) + 8/8 \times \log_2(8/8)]$$

$$= - [0/8 \times (?) + 8/8 \times (0)]$$

$$= - [0]$$

$$= 0$$

(Minimum impurity) – not good for training

Conclusion: If a dataset can be divided into classes then the dataset is non-homogenous and impurity (Entropy) is high.

Example 2:

- Entropy(9+, 7-)
= Work out the values

- Log base 2 calculator
- <https://www.calculator.net/log-calculator.html?xv=8&base=2&yv=&x=79&y=21>
- <https://www.math.unipd.it/~aiolli/corsi/0708/IR/Lez12.pdf>

$$\text{entropy}(\text{Set}) = I(\text{Set}) = -\sum_{i=1}^k P(\text{value}_i) \cdot \log_2(P(\text{value}_i))$$

<u>Color</u>	<u>Size</u>	<u>Shape</u>	<u>Edible?</u>
Yellow	Small	Round	+
Yellow	Small	Round	-
Green	Small	Irregular	+
Green	Large	Irregular	-
Yellow	Large	Round	+
Yellow	Small	Round	+
Yellow	Small	Round	+
Yellow	Small	Round	+
Green	Small	Round	-
Yellow	Large	Round	-
Yellow	Large	Round	+
Yellow	Large	Round	-
Yellow	Large	Round	-
Yellow	Large	Round	-
Yellow	Small	Irregular	+
Yellow	Large	Irregular	+

• Entropy(9+, 7-)

$$= - [P(+) \times \log_2(P(+)) + P(-) \times \log_2(P(-))]$$

$$= - [9/16 \times \log_2(9/16) + 7/16 \times \log_2(7/16)]$$

$$= - [9/16 \times (\log_2 0.5625) + 7/16 \times (\log_2 0.4375)]$$

$$= -[0.5625 \times (-0.8301) + 0.4375 \times (-1.1926)]$$

$$= +0.9887$$

Max Entropy Values

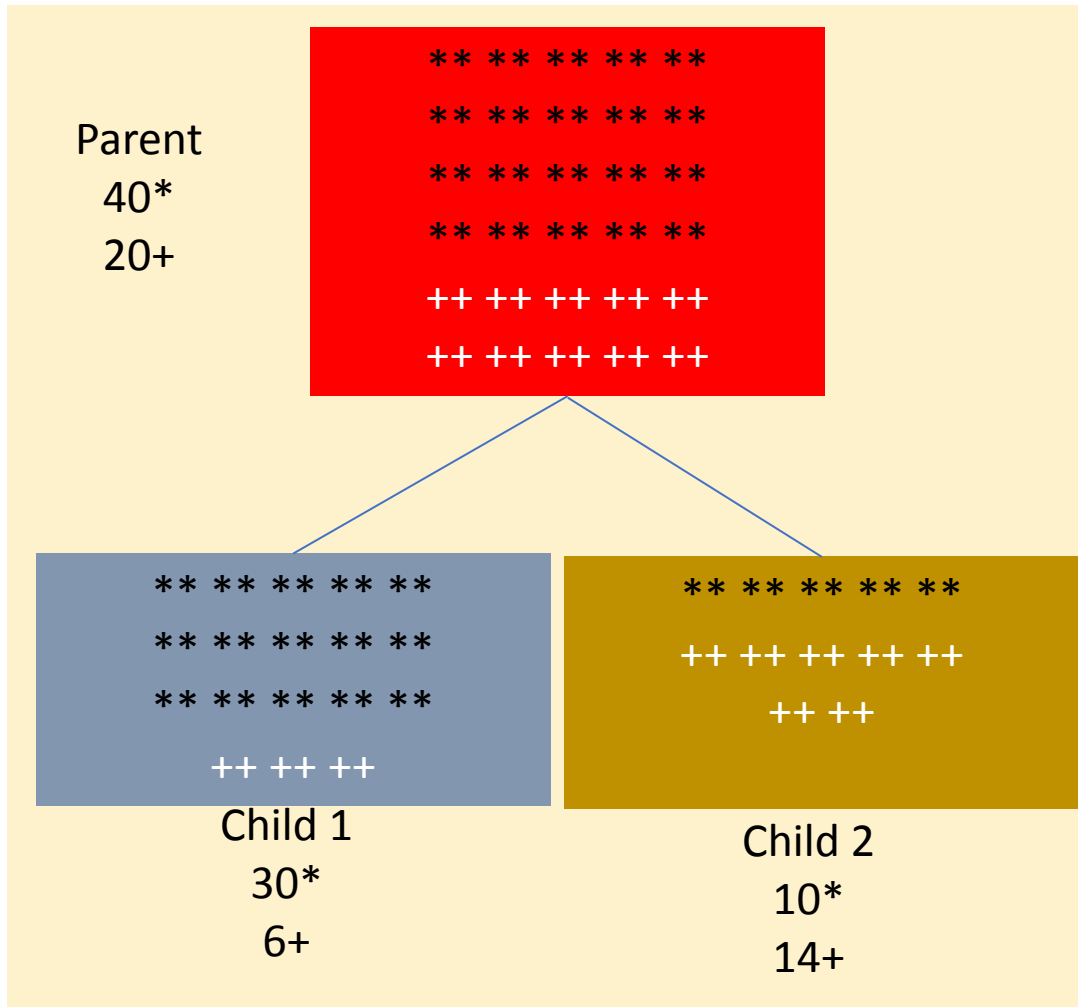
- Two classes: Max entropy is 1.
- Four Classes: Max entropy is 2.
- Eight Classes: Max entropy is 3.
- 16 classes: Max entropy is 4.

Information Gain

- We want to determine which attribute in a given set of training feature vectors is most useful for discriminating between the classes to be learned.
 - Information Gain is used in feature selection
 - Information Gain tells us how important a given attribute of the feature vectors is
 - Constructing a **decision tree** is all about finding an attribute that returns the **highest information gain** and the **lowest possible entropy**.
-
- <https://homes.cs.washington.edu/~shapiro/EE596/notes/InfoGain.pdf>
 - <https://medium.com/analytics-vidhya/entropy-calculation-information-gain-decision-tree-learning-771325d16f>

Information Gain Formula

$$\text{Information Gain} = \text{entropy}(\text{parent}) - [\text{weighted entropy}(\text{children})]$$



$$\begin{aligned}\text{Parent Entropy: Entropy}(40^*, 20^+) &= -[40/60 \times \log_2(40/60) + 20/60 \times \log_2(20/60)] \\ &= -[40/60 \times (-0.584) + 20/60 \times (-1.586)] \\ &= -[-0.919] \\ &= 0.919\end{aligned}$$

$$\begin{aligned}\text{Child 1 Entropy: Entropy}(30^*, 6^+) &= -[30/36 \times \log_2(30/36) + 6/36 \times \log_2(6/36)] \\ &= -[30/36 \times (-0.264) + 6/36 \times (-2.582)] \\ &= -[-0.651] \\ &= 0.651\end{aligned}$$

$$\begin{aligned}\text{Child 2 Entropy: Entropy}(10^*, 14^+) &= -[10/24 \times \log_2(10/24) + 14/24 \times \log_2(14/24)] \\ &= -[10/24 \times (-1.279) + 14/24 \times (-0.778)] \\ &= -[-0.991] \\ &= 0.991\end{aligned}$$

$$\begin{aligned}\text{Information Gain} &= 0.919 - 36/60 \times 0.651 - 24/60 \times 0.991 \\ &= 0.132\end{aligned}$$

Look at Complete Example for Calculation of Information Gain

$$G(S, A) = I(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} I(S_v)$$

- S – Set
- A – Attribute (i.e. Feature, Vector)
- $\text{Values}(A)$ – all values of Attribute (i.e. Feature, Vector)
- S_v – subset of S whereby attribute A has value v
- <https://medium.com/analytics-vidhya/entropy-calculation-information-gain-decision-tree-learning-771325d16f>

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Figure 1: Dataset of playing tennis, which will be used for training decision tree

- Attribute 1 – Wind (values are weak, strong)
- Attribute 2 – Humidity (values are high, normal)
- Attribute 3 – Temperature (values are hot, mild, cool)
- Attribute 4 – Outlook (values are sunny, overcast, rain)

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Figure 1: Dataset of playing tennis, which will be used for training decision tree

- Calculate the information gain for attribute Wind

- $S = [9 \text{ yes}, 5 \text{ No}]$

- $S_{\text{weak}} = [6 \text{ yes}, 2 \text{ No}]$

- $S_{\text{strong}} = [3 \text{ yes}, 3 \text{ No}]$

- $\text{Entropy}(S) = 0.940$

- $\text{Entropy}(S_{\text{weak}}) = - [6/8 \times \log_2(6/8) + 2/8 \times \log_2(2/8)] = 0.811$

- $\text{Entropy}(S_{\text{strong}}) = - [3/6 \times \log_2(3/6) + 3/6 \times \log_2(3/6)] = 1.0$

- **Information Gain(S, Wind)**

$$= \text{Entropy}(S) - [8/14 \times \text{Entropy}(S_{\text{weak}}) + 6/14 \times \text{Entropy}(S_{\text{strong}})]$$

$$= 0.940 - 8/14 \times 0.811 - 6/14 \times 1.0$$

$$= 0.940 - 0.463 - 0.429$$

$$= 0.048$$

Calculated Information Gain values for all the 4 attributes are:

- $\text{Gain}(S, \text{Wind}) = 0.048$
 - $\text{Gain}(S, \text{Outlook}) = 0.246$
 - $\text{Gain}(S, \text{Humidity}) = 0.151$
 - $\text{Gain}(S, \text{Temperature}) = 0.029$
-
- The main goal of information gain is to find the attribute that is most useful for classifying the training dataset.
 - Outlook will be the root of the tree

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Figure 1: Dataset of playing tennis, which will be used for training decision tree

- Outlook = sunny (D1, D2, D8, D9, D11); (2 yes, 3 no)
- Outlook = overcast (D3, D7, D12, D13); (4 yes, 0 no) -> yes
- Outlook = rain (D4, D5, D6, D10, D14); (3 yes, 2 no)

- <https://medium.com/analytics-vidhya/entropy-calculation-information-gain-decision-tree-learning-771325d16f>

- Do leaf by leaf
- Sunny Leaf

$$S_{\text{sunny}} = 5$$

Humidity_{high} = [0 yes, 3 No] -> No
 Humidity_{normal} = [2 yes, 0 no] -> Yes

$$\begin{aligned} \text{Gain}(S_{\text{sunny}}, \text{Humidity}) &= \text{Entropy}(S_{\text{sunny}}) - [3/5 \times \text{Entropy}(\text{Humidity}_{\text{high}}) \\ &+ 2/5 \times \text{Entropy}(\text{Humidity}_{\text{normal}})] \\ &= 0.970 \end{aligned}$$

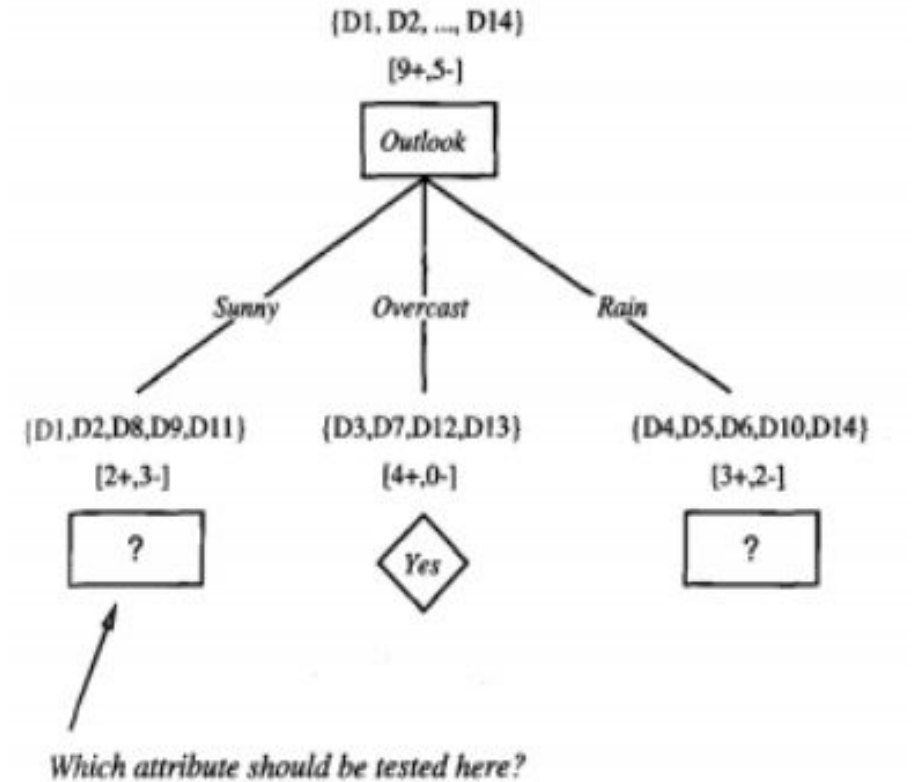
Results

$$\begin{aligned} \text{Gain}(S_{\text{sunny}}, \text{Humidity}) &= 0.970 \\ \text{Gain}(S_{\text{sunny}}, \text{Temperature}) &= 0.570 \\ \text{Gain}(S_{\text{sunny}}, \text{Wind}) &= 0.019 \end{aligned}$$

Conclusion

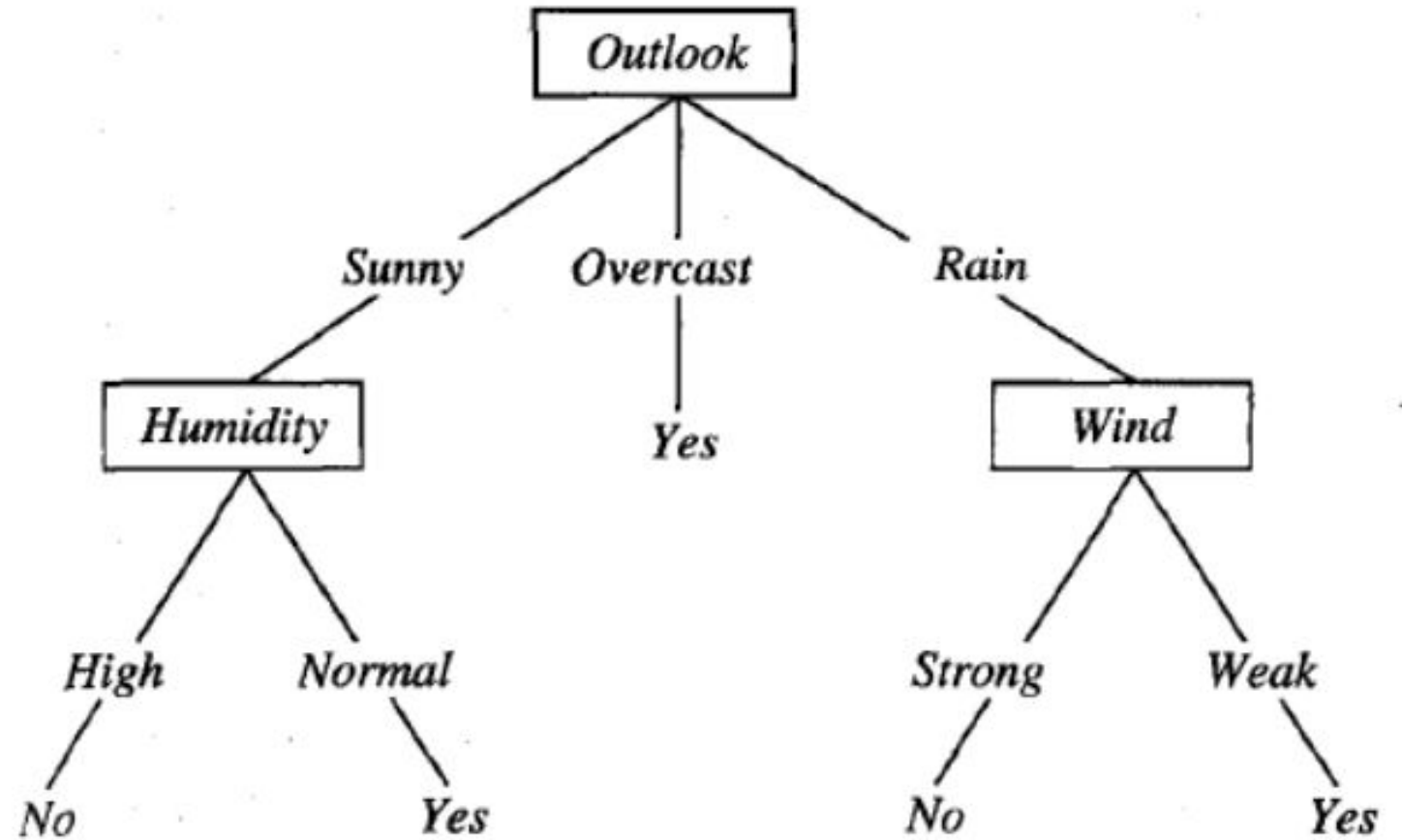
Child node for Outlook = sunny is Humidity

The leaf for Humidity_{high} is No
 The leaf for is Humidity_{normal} is Yes
 Both cannot be further partitioned



Day	Outlook	Temp	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes

- Repeat for Outlook = Rain



Gini Index

- **Gini index** or **Gini** impurity measures the degree or probability of a particular variable being wrongly classified when it is randomly chosen
- **Gini Impurity** tells us what is the probability of misclassifying an observation.
- The **lower the Gini the better** the split.
- A **Gini Index** of 0.5 denotes equally distributed elements into some classes.
- Favours larger partitions
- The algorithm works as $1 - [P(\text{class1})^2 + P(\text{class2})^2 + \dots + P(\text{classN})^2]$

Gini Index Formula

$$Gini = 1 - \sum_j p_j^2$$

- **Gini Index** = deducting the sum of squared of probabilities of each class from 1
- P_i – denoted the probability of an element being classified for a distinct class
- Example – Look at this <https://blog.quantinsti.com/gini-index/>

Note: The Column Return is the Target/Dependent Variable

Calculate Gini Index for Past Trend

- $P(\text{Past Trend}=\text{Positive}): 6/10$
- $P(\text{Past Trend}=\text{Negative}): 4/10$

Past Trend = Positive

- If (Past Trend = Positive & Return = Up), probability = $4/6$
- If (Past Trend = Positive & Return = Down), probability = $2/6$
- $\text{Gini index} = 1 - ((4/6)^2 + (2/6)^2) = 0.45$

Past Trend = Negative

- If (Past Trend = Negative & Return = Up), probability = $0/4$
- If (Past Trend = Negative & Return = Down), probability = $4/4$
- $\text{Gini index} = 1 - ((0)^2 + (4/4)^2) = 0$

Overall Past Trend (Positive and Negative)

- Weighted sum of the Gini Indices can be calculated as follows:
- $\text{Gini Index for Past Trend} = (6/10) \times 0.45 + (4/10) \times 0 = 0.27$

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

Calculation of Gini Index for Open Interest

- $P(\text{Open Interest}=\text{High}): 4/10$
- $P(\text{Open Interest}=\text{Low}): 6/10$

For Open Interest = High

- If (Open Interest = High & Return = Up), probability = $2/4$
- If (Open Interest = High & Return = Down), probability = $2/4$
- $\text{Gini index} = 1 - ((2/4)^2 + (2/4)^2) = 0.5$

For Open Interest = Low

- If (Open Interest = Low & Return = Up), probability = $2/6$
- If (Open Interest = Low & Return = Down), probability = $4/6$
- $\text{Gini index} = 1 - ((2/6)^2 + (4/6)^2) = 0.45$

For Overall Open Interest (High and Low)

- Weighted sum of the Gini Indices can be calculated as follows:
- $\text{Gini Index for Open Interest} = (4/10) \times 0.5 + (6/10) \times 0.45 = 0.47$

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

Repeat for Trading Volume

Results

Attributes/Features	Gini Index
Past Trend	0.27
Open Interest	0.47
Trading Volume	0.34

Lowest Gini Index

Past Trend – will be used as Root of the tree

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

Calculate the Gini Index for the Positive Branch

See

<https://blog.quantinsti.com/gini-index/>

- Calculate Gini Index of Open Interest for Positive Past Trend
- Calculate Gini Index of Trading Volume for Positive Past Trend
- Results
- Trading Volume has the minimum Gini Index and it will be the next node.

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Positive	Low	High	Up
Positive	High	High	Up
Positive	Low	Low	Down
Positive	Low	Low	Down
Positive	High	High	Up

Attribute/Feature	Gini Index
Open Interest	0.33
Trading Volume	0.00

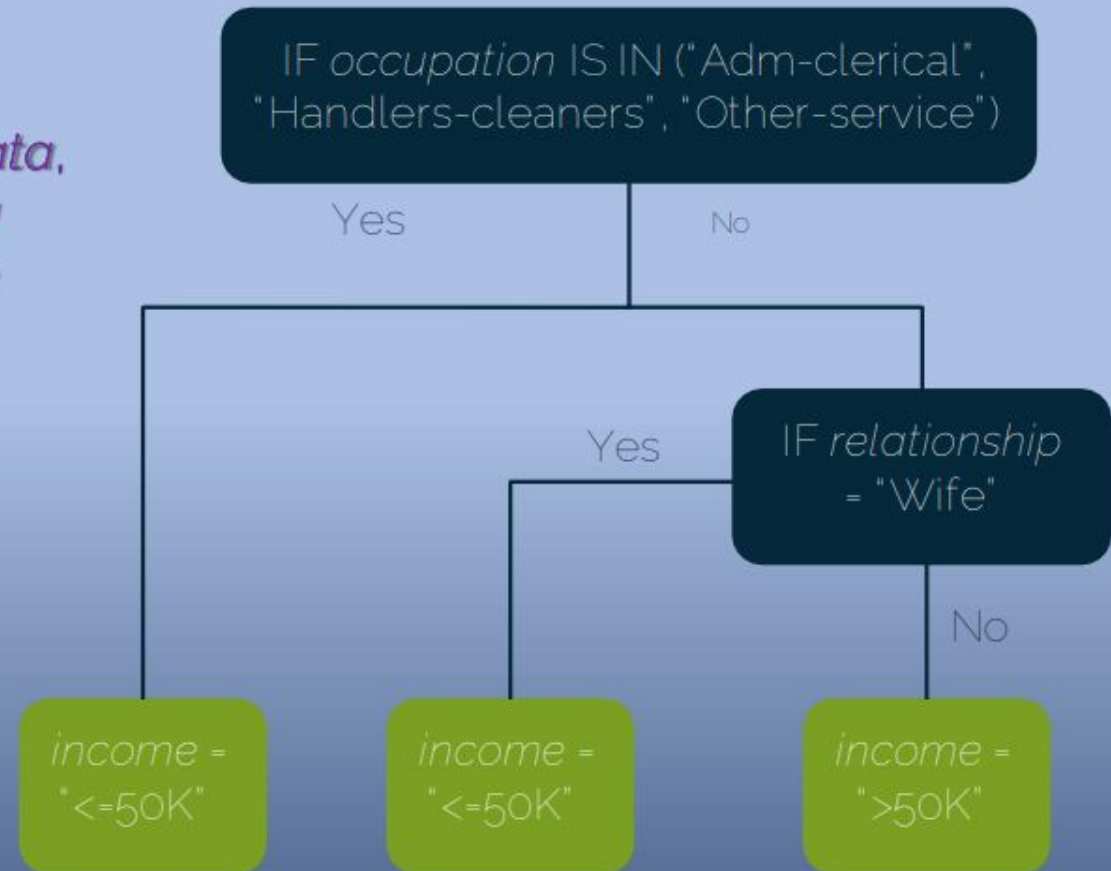
Overfitting

- Error rate in predicting the correct class for new cases
 - overfitting of test data
 - very low apparent error rate
 - high actual error rate

The Overfitting Problem of Machine Learning

	occupation	relationship	income
1	Adm-clerical	Not-in-family	<=50K
2	Exec-managerial	Husband	<=50K
3	Handlers-cleaners	Not-in-family	<=50K
4	Handlers-cleaners	Husband	<=50K
5	Prof-specialty	Wife	<=50K
6	Exec-managerial	Wife	<=50K
7	Other-service	Not-in-family	<=50K
8	Exec-managerial	Husband	>50K
9	Prof-specialty	Not-in-family	>50K
10	Exec-managerial	Husband	>50K

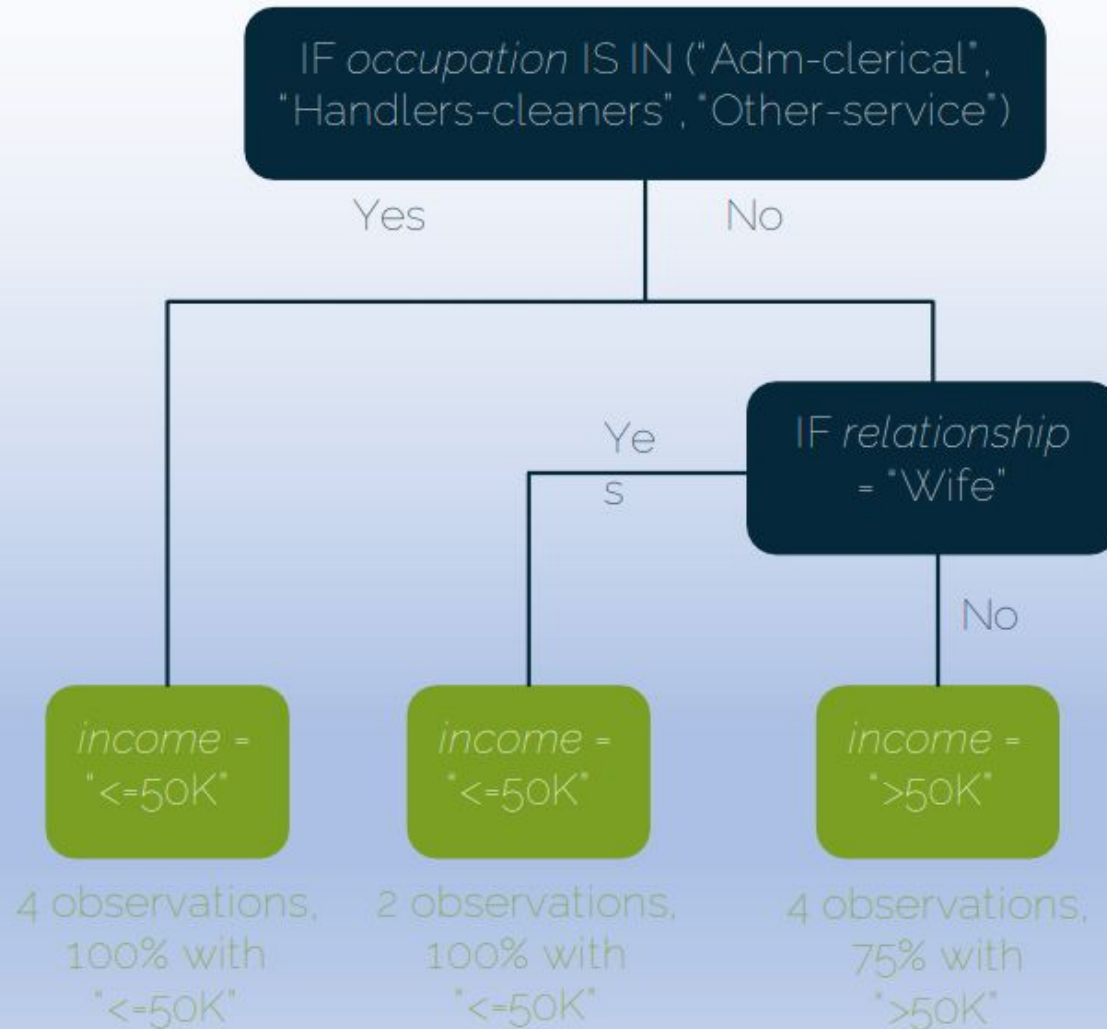
The combination of data, algorithm, and training regimen produced the following model...



The Model Is Good!

Considering the data used to *train* the model, things look awesome!

	occupation	relationship	income
1	Adm-clerical	Not-in-family	<=50K
2	Exec-managerial	Husband	<=50K
3	Handlers-cleaners	Not-in-family	<=50K
4	Handlers-cleaners	Husband	<=50K
5	Prof-specialty	Wife	<=50K
6	Exec-managerial	Wife	<=50K
7	Other-service	Not-in-family	<=50K
8	Exec-managerial	Husband	>50K
9	Prof-specialty	Not-in-family	>50K
10	Exec-managerial	Husband	>50K

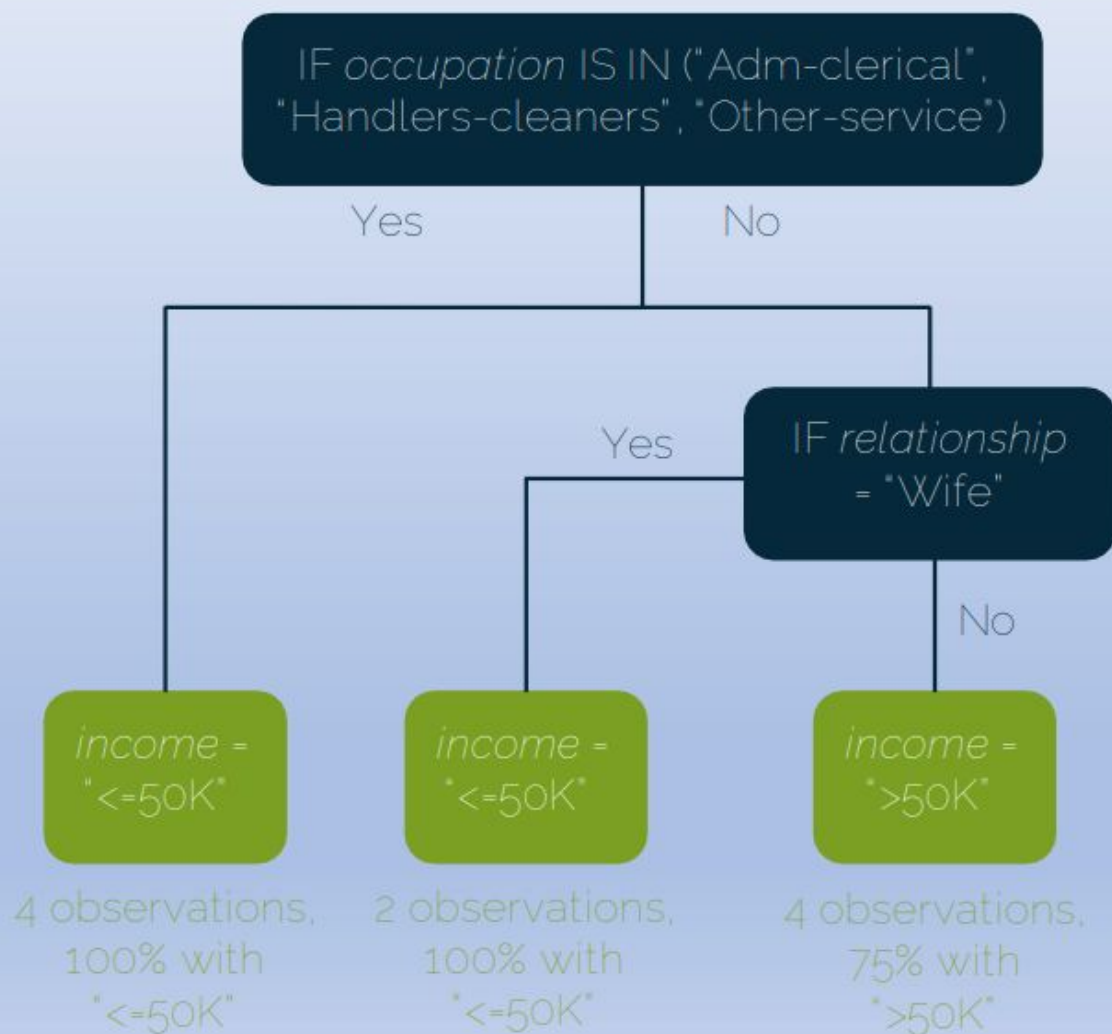


This model gets 9 out of 10 (i.e., 90%) of labels correct!

Or Is It?

	occupation	relationship	income
1	Prof-specialty	Wife	>50K
2	Adm-clerical	Wife	>50K
3	Exec-managerial	Wife	>50K
4	Other-service	Husband	>50K
5	Other-service	Husband	>50K
6	Other-service	Wife	>50K
7	Exec-managerial	Husband	<=50K
8	Sales	Not-in-family	<=50K
9	Transport-moving	Husband	<=50K

This is the
essence of
overfitting



This model
gets 9 out of 9
(i.e., 100%) of
labels wrong!

What Happened?

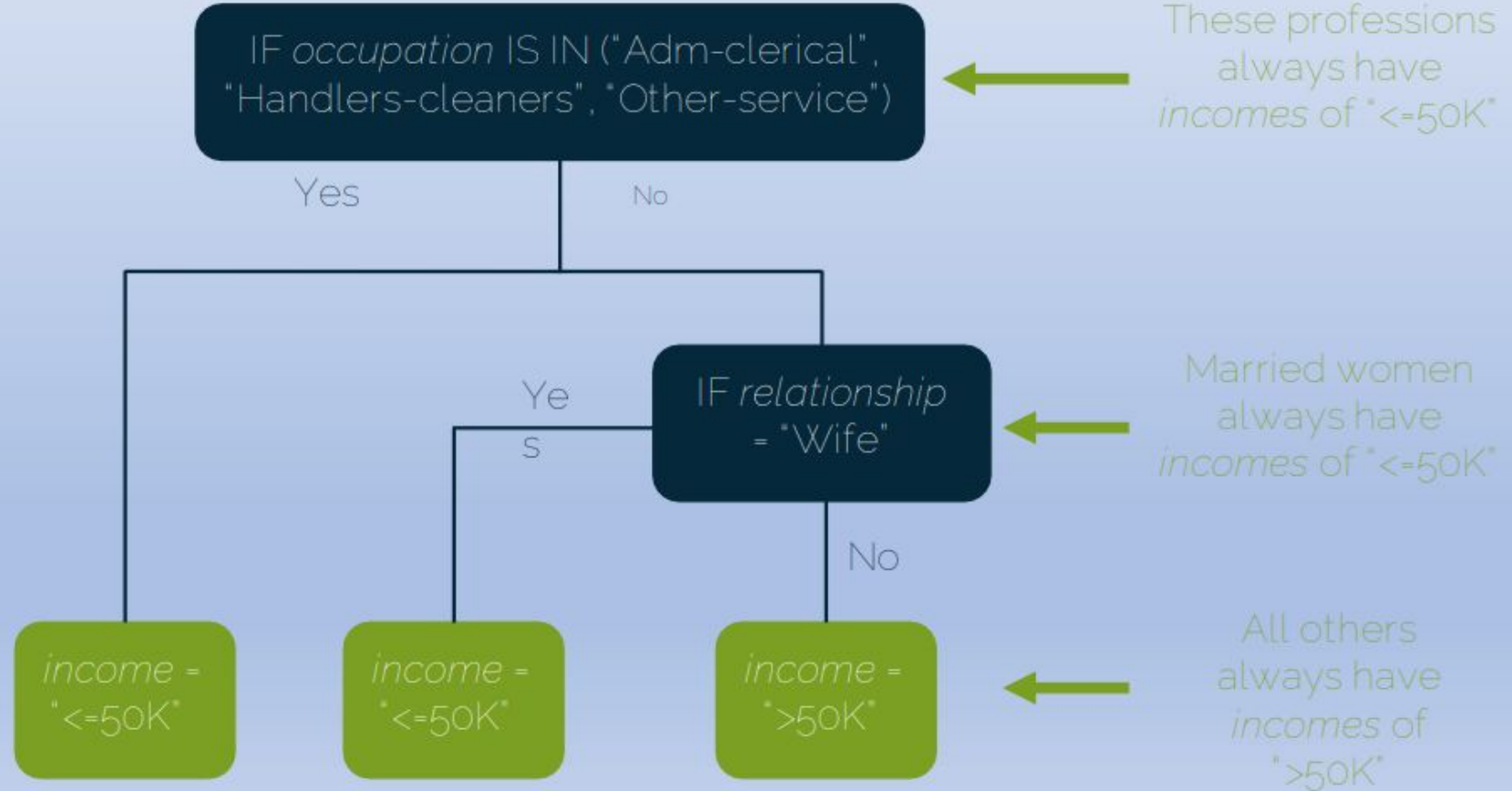
In this example, the overfitting can be blamed on the *training regimen*...

Given this small amount of data...

	occupation	relationship	income
1	Adm-clerical	Not-in-family	<=50K
2	Exec-managerial	Husband	<=50K
3	Handlers-cleaners	Not-in-family	<=50K
4	Handlers-cleaners	Husband	<=50K
5	Prof-specialty	Wife	<=50K
6	Exec-managerial	Wife	<=50K
7	Other-service	Not-in-family	<=50K
8	Exec-managerial	Husband	>50K
9	Prof-specialty	Not-in-family	>50K
10	Exec-managerial	Husband	>50K

The greediness of decision trees makes them prone to overfit!

This model is far too specialized (or *complex*)!



Decision Tree Limitations

- No backtracking
 - local optimal solution not global optimal solution
 - *lookahead* features may give us better trees
- Rectangular-shaped geometric regions
 - in two-dimensional space
 - regions bounded by lines parallel to the x- and y- axes
 - some linear relationships not parallel to the axes

Conclusions

■ Utility

- analyze classified data
- produce
- accurate and easily understood classification rules
- with good predictive value

Improvements

- Limitations being addressed
- multivariate discrimination - oblique trees
- data mining techniques

Resource

Harvard Business Review

- <https://hbr.org/1964/07/decision-trees-for-decision-making>

Model Evaluation

<http://www.learnbymarketing.com/481/decision-tree-flavors-gini-info-gain/>

<https://homes.cs.washington.edu/~shapiro/EE596/notes/InfoGain.pdf>

End